21. In *E.coli*, three enzymes β- galactosidase, permease and transacetylase are produced in the presence of lactose. Explain why the enzymes are not synthesized in the absence of lactose.

22. Distinguish between structural gene, regulatory gene and operator gene.

23. A low level of expression of *lac* operon occurs at all the time. Justify the statement.

24. HGP is the windows for treatment of various genetic disorders. Justify the statement.

25. Why the human genome project is called a mega project?

26. From their examination of the structure of DNA, What did Watson and Crick infer about the probable mechanism of DNA replication, coding capability and mutation?

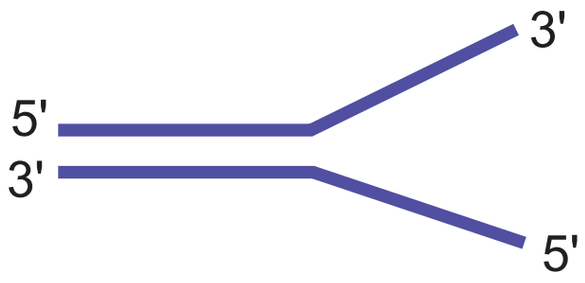
27. Why tRNA is called an adapter molecule?

28. What are the three structural differences between RNA and DNA?

29. Name the anticodon required to recognize the following codons: AAU, CGA, UAU, and GCA.

30. a) Identify the figure given below

    b) Redraw the structure as a replicating fork and label the parts



    c) Write the source of energy for this replication and name the enzyme involved in this process.

    d) Mention the differences in the synthesis of protein, based on the polarity of the two template strands.

31. If the coding sequence in a transcription unit is written as follows:

    5' TGCATGCATGCATGCATGCATGCATGC 3'
    Write down the sequence of mRNA.

32. How is the two stage process of protein synthesis advantageous?

33. Why did Hershey and Chase use radioactively labelled phosphorous and sulphur only? Would they have got the same result if they use radiolabelled carbon and nitrogen?

34. Explain the formation of a nucleosome.

35. It is established that RNA is the first genetic material. Justify giving reasons.

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